

Multiple Choice (10 marks)

1. Electromagnetic radiation emitted from a nucleus is most likely to be in the form of
 - (a) gamma rays
 - (b) microwaves
 - (c) ultraviolet radiation
 - (d) visible light
2. An atom has filled $n = 1$ and $n = 2$ levels. How many electrons does the atom have?
 - (a) 2
 - (b) 4
 - (c) 6
 - (d) 10
3. A resistor in a circuit dissipates energy at a rate of 1 W. If the voltage across the resistor is doubled, what will be the new rate of energy dissipation?
 - (a) 0.25 W
 - (b) 0.5 W
 - (c) 1 W
 - (d) 4 W
4. Which of the following statements about bosons and/or fermions is true?
 - (a) Bosons have symmetric wave functions and obey the Pauli exclusion principle.
 - (b) Bosons have antisymmetric wave functions and do not obey the Pauli exclusion principle.
 - (c) Fermions have symmetric wave functions and obey the Pauli exclusion principle.
 - (d) Fermions have antisymmetric wave functions and obey the Pauli exclusion principle.
5. The rest mass of a particle with total energy 5.0 GeV and momentum 4.9 GeV/c is approximately
 - (a) $0.1 \text{ GeV}/c^2$
 - (b) $0.2 \text{ GeV}/c^2$
 - (c) $0.5 \text{ GeV}/c^2$
 - (d) $1.0 \text{ GeV}/c^2$

True / False (10 marks)

Answer True or False

6. True or False: A helium-neon laser is the most versatile laser for performing spectroscopy across visible wavelengths.
7. True or False: The negative muon (μ^-) has properties that are most similar to those of a meson.
7. True or False: The angular magnification of a refracting telescope with an eye-piece lens of 20 cm focal length is 6.
8. True or False: Phosphorus is a suitable dopant for germanium to form an n-type semiconductor.
9. True or False: Gas lasers operate by utilizing transitions between the energy levels of free atoms in a gaseous medium.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the behavior of unpolarized light as it passes through two ideal linear polarizers oriented at a 45-degree angle to each other. Calculate the transmitted intensity as a percentage of the incident intensity and explain the underlying principles involved.
12. Discuss the distribution of kinetic energy in a uniform solid disk rolling down an inclined plane without slipping. What fraction of its total kinetic energy is rotational, and why?

End of Paper